Exercise 2: Sabre Dance Answers

Difficulty Rating ★★☆☆☆

MACS Document Team

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Task 1: Fundamental Feature Identification

Feature	Rule	Cruise (14 kt)	High Speed (28 kt)
Shaft Rate (SR)	RPM / 60	1.4	2.8
Blade Rate (BR)	$4 \times SR$	5.6	11.2
Blade : Shaft	Invariant	4 : 1	4 : 1
Crankshaft (CSR)	$6 \times SR$	8.4	16.8
Gearing Ratio (GR)	CSR/SR	6 : 1	6 : 1
Engine Firing Rate (EFR)	$CSR \times \frac{N_{cylinders}}{2}$	84	168
EFR Harmonics	2x, 3x ...	168, 252	336, 504
Half-order picket	$3 \times SR$	Every 4.2	Every 8.4
Gear-mesh (GMF)	$45 \times SR$	63.0	126.0
Mains (fixed)	Genset	50 (+100, 150)	50 ...
Genset firing (fixed)	Genset	125	125

Task 2: LOFARgram

Ensure all relevant tonals for 14 and 28 kts are marked. Check that 50 hz electrical mains and 125 hz Genset firing remain fixed for both speeds. For extra marks check harmonics are included and properly annotated. Finally for top marks ensure sidebands on the gear-mesh tonal are included.

Task 3: Effect of speed and direction change

1. Changes in speed causes key engine and propeller components to rotate faster increasing the frequency.
2. Mains and Genset remain constant during speed changes as supply remains constant. Sudden frequency changes would damage delicate electrical equipment.
3. Doppler effect cause by changes in relative motion of the sound source.
4. Doppler affects all sound sources and is independent of the actual sound source. It is caused by the stretching and compressing of the sound produced by the vessel.

Task 4: Speed Estimate

The significant conclusion from this exercise is that although a good enough estimate of vessel speed can be obtained from SONAR it is only part of the process. Target Motion Analysis is also part of the process. The point of this task is to reinforce that **uncertainty** is part of day to day life of a SONAR operator. Assessing the balance of probabilities is a key skill.

Task 5: Vessel Matching and Classification

1. Mystery01 - Fail or partial - this is not Sabre. Note: Turbine howl at 945 Hz is a distinctive discriminator.
2. Mystery02 - Pass or partial - this is Sabre

Practice, practice, practice. Use the new Training tool in MACS to practice matching and classifying vessels. Again the key take away here is to learn to live with uncertainty.